U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

Test Operations Procedure 04-2-831A 23 July 2025

USE OF BLAST TEST DEVICE (BTD) DURING BLAST OVERPRESSURE MEASUREMENT

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*This TOP supersedes 04-2-831, Use of Blast Test Device (BTD) During Auditory Blast Overpressure Measurement, 12-08-2008

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1. SCOPE.

The objective of this Test Operations Procedure (TOP) is to familiarize the reader with the Blast Test Device (BTD) and its use in blast overpressure testing. A BTD is a rigid cylindrical test apparatus approximating a human torso and can be used in blast overpressure tests to record the auditory and non-auditory health hazards of a weapon or explosive.

A fully assembled BTD can be seen in Figure 1 below.



Figure 1 – Fully Assembled BTD

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2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

Item	Requirement
Firing Range	A firing range must be selected to meet test requirements and to provide adequate protection for personnel and equipment. Firing points must be sufficiently protected and remotely located to safely contain a premature ammunition detonation, ricochet, or a catastrophic weapon failure as well as contain discarding parts from projectiles.
Test Facility	For free-field static blasts, the test facility shall have uniform grade free of all sound reflecting surfaces within 30meters(m) (98feet(ft)) of the test item and measuring equipment, if practical. Depending on the desired test scenario, the BTD may be used in enclosed spaces.

2.2 Instrumentation.

The thorax pressure transducers can be either piezo-resistive or piezo-electric and must have at least a 10 kilohertz(kHz) frequency response after filtering. The auditory sensor must be piezoelectric and have at least 40 kHz frequency response after filtering. Rise times for these filter frequencies are 9 microseconds(μs) and 35μs, respectively.

The U.S. Army Aberdeen Test Center employs the Endevco model 8530C piezoresistive pressure transducer for thoracic pressure measurements. This model sensor is available in the range of 15 to 100 psia. For applications requiring higher pressure thresholds, the Endevco model 8530B is available. It is essential to select the appropriate gauge model and pressure range to align with the anticipated pressure levels for accurate measurements.

The PCB Piezotronics model 102M114 sensor is employed for precise auditory measurements. This fully threaded design, devoid of a shoulder, facilitates seamless integration into its adapter housing. Additionally, it features a specialized ablative coating that provides effective thermal protection, ensuring optimal performance in varying thermal conditions.

ATC conducts calibration of the two aforementioned pressure transducers utilizing a low-level dynamic pressure calibration fixture, model K9903C, manufactured by The Modal Shop, a wholly

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owned subsidiary of PCB Piezotronics, Inc. The calibration of the sensors is performed in-house on an annual basis and are traceable to the National Institute of Standards and Technology (NIST). A tolerance of ±5% from previous calibration and ±7% from the original established baseline are permissible. Deviations exceeding these thresholds are classified as a failure. There is an estimated calibration uncertainty of ± 1.0% at a 95% confidence level. During the calibration process, ambient conditions are approximately 20°C with a relative humidity of 40%.

If a preference for sensors from an alternative model or manufacturer arises, the following guidelines should be followed when making selection:

Devices for Measuring	Permissible Measurement Uncertainty
Pressure gauge	
Uncertainty	As required
Resonance frequency	Minimum 100 kHz, 400 kHz or larger desirable
Linearity	< 3% Full scale output (FSO)
Acceleration Sensitivity (axial)	Less than 1.5 pascal ($Pa/m/s^2$)
Acceleration Sensitivity (transverse)	Less than 7 $Pa/m/s^2$
Overall System Linearity	<3%
Electronic recording system	±1%
Low Pass Filtering	
Auditory Transducers	40 kHz, 6-pole, Bessel
Non-Auditory Transducers	10 kHz, 6-pole, Bessel
Air pressure	±0.3%
Temperature	±0.07%/°C
Wind speed and direction	±0.3 meters per second(m/s) (± 0.1 feet per second(ft/s)) ± 1.6 degrees (°)

2.3 Data Acquisition and Equipment

Data acquisition instrumentation must meet or exceed the recording instrumentation specifications provided in Military Standard (MIL-STD) 810H, Method 522.2 Ballistic Shock, Section 4.4.2.

3. REQUIRED TEST CONDITIONS.

Wind speeds shall not exceed 5 m/s (16.4 ft/s). Conditions may allow for higher wind speeds, provided that the steady state noise induced by the wind is 40 decibels (dB) below the maximum overpressure expected, and the increase in overpressure on the downwind side, caused by apparent shortening of the path length between the source and the sensor, is less than 2 dB.

The firing range selected should meet test requirements to provide adequate protection for personnel and equipment; firing points must be sufficiently protected and remotely located to safely contain a premature ammunition detonation, a ricochet, or a catastrophic weapon failure as well as contain discarding parts from projectiles.

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For free field testing, the test facility will be uniform grade free of all sound reflecting surfaces within 30m (100 ft) of the test item and measuring equipment, if practical.

Prior to the use of each gage in outlined test conditions, each gage must be calibrated in accordance with NIST in a laboratory in accordance with ANSI/ISO/IEC 17025:2017. Should a gage be calibrated, verified or certified outside of ANSI/ISO/IEC 17025:2017, a waiver may be accepted by knowledgeable individuals who have the associated knowledge of NIST calibration and the proposed use of the gage. Failure to follow ANSI/ISO/IEC 17025:2017 could result in failure of a test and must be repeated with gages in accordance with National Standard.

Field testing of the entire acquisition system in front of dynamic pressure by means of explosions (or equivalent) is desired. An arena calibration test for BOP should be performed before testing to ascertain transducer/gauge operation and reliability. An arena with a known charge type and weight should be used. The area should be as flat as possible with no reflective surfaces. Transducers should be placed in the arena at a specific distance from a center located explosive charge as well as equidistant from each other around the circumference. Several extra positions will be recorded to account for errors that can occur with transducers under this dynamic testing environment. This will enable hand picking the best wave conformation channel sets (gauge, cable, connections and channel used) recorded.

Make sure transducers are isolated from ground, shock-mounted, flash/thermally protected, and operated within the specified temperature and pressure ranges. Place recording instrumentation far enough away from the detonation to be minimally affected by the blast. Make sure maximum cable length from the instrumentation point to most remote gauge does not inhibit the frequency response of that gauge. When long lines are required, line drivers are needed to reduce outside inference, to maintain required frequency response.

When taking measurements to compare data with national standards for human tolerance to muzzle blasts, other test requirements should not jeopardize these test results.

NOTE: Give careful attention to test setup and data analysis to ensure that results are representative of the blast wave, rather than displaying anomalies due to extraneous effects. The measurement situation will determine the type of transducer and mounting arrangement.

4. BLAST TEST DEVICE.

4.1 Background

Blast Overpressure Health Hazard Assessment (BOP-HHA) testing is required for fielding weapon systems that produce blast overpressure levels that can potentially result in auditory and non-auditory (blast lung) injuries. The current standard software for assessing nonauditory blast lung injury is INJURY 8.3, released by the U.S. Army Medical Research and Materiel Command (USAMRMC). (REFERENCE 4). INJURY 8.3 requires the Blast Test Device (BTD) to collect pressure loading data for the four sides of the thorax as inputs to the model.

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For occupational lung injury health hazard assessment, Defense Health Agency- Public Health (DHA-PH) uses a customized version of the INJURY model, known as BOP-HHA Version 2.0, to conduct health hazard assessment for cumulative daily exposure to blast overpressures. Based on acceptable injury risk levels and risk tolerances, permissible daily number of exposures can be computed for different hazard severity assessment levels. This provides the basis for setting the weapon's daily allowable number of rounds for weapons training.

Because of flow blockage effects, it is essential to use the BTD to collect pressure loadings on the thorax, as opposed to just using an open field pencil gauge to measure incident pressure. An integral part of the validated INJURY methodology, the BTD, is the standard device for collecting blast overpressure data and has been used extensively by the testing community for many years – for both occupational and survivable blast testing.

4.2 History

Although the original design height for the BTD was 30 inches(in.) (developed by L3 Applied Technologies, formerly Jaycor), several different heights have been used throughout the years. BTDs of heights 6, 12, 24 and 30 in. have been used in special circumstances to accommodate certain weapon configurations and test requirements (see Figure 2) that larger BTDs were too big to support. It is strongly advised that the 6-in. and 12-in. BTDs cannot be employed, as they have not undergone the necessary evaluation and approval processes for use.

**The use of brand names does not constitute endorsement by the Army or any other agency of the Federal Government, nor does it imply that it is best suited for its intended application.



Figure 2 – Small BTD Used for Prone Firing Position

The original BTD version has thorax transducers located at 15 in. from the top of the BTD. However, this is not consistent with torso measurements of the 50th percentile male. Because of this, ATC developed a 24-in. and 30-in. version that has the transducers located 6 in. and 9 in., respectively, from the top of the BTD, allowing for a chest to ear measurement that more closely matches that of a 50th percentile male solider. The 24-in. BTD is a more reasonable approximation of the crotch to shoulder dimensions of a 50th percentile male and offers greater flexibility in positioning the BTD in various test configurations. The chest to ear measurement is the same in the 24- and 30-in. BTDs.

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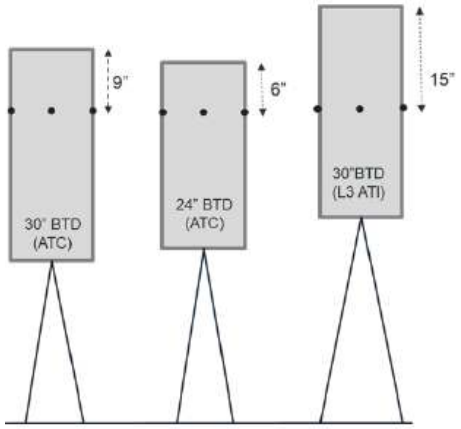


Figure 3 – Comparison of Three BTD Types

Recently, in 2021, testing was conducted to compare the 24-in. BTD to the ATC 30-in. and original 30-in. BTD designs (Figure 3). An extensive series of open air, compartment tests and shock tube testing were conducted in a joint effort by the U.S. Army Aberdeen Test Center (ATC), DHA-PH and L3 Applied Technologies. This testing helped to reveal the differences in thoracic response between the three variants of BTD and provide updates to the BOP Health Hazard Assessment (HHA) model for each. The updated BOP HHA model now provides a dropdown selection where the user can select the BTD being used for testing. Please note that the Army Public Health Command exclusively possesses the BOP Health Hazard Software; therefore, all analyses must be conducted by this authority.

Caution must be used when running the accompanying injury analysis software to predict thoracic injury; the software was designed to accommodate 24- and 30-in. BTDs only. Although 12 and 6 in. BTDs have historically been used for crouching, sitting, kneeling positions when using shoulder fired Anti-Tank Weapons or placement in certain combat vehicle seats, they should no longer be used in this type of testing.

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4.3 Overview

The BTD, shown in Figures 3 and 4, is an aluminum cylinder, with a 12 in. outside diameter, a 0.25-in. wall thickness, and is either 24 or 30 in. tall. Two, 0.75 in. thick, stiffener disks are fastened to the interior of the cylinder, one 5.5 in. from the bottom and one 5.5 in. from the top for the ATC 30 in BTD. The original 30-in. BTD has stiffener disks flush with the top and bottom of the BTD. Stiffener disks are located at 6 in. from the bottom and 3 in. from the top for the ATC 24-in. BTD.

Four piezo-resistive or piezo-electric transducers are mounted on the cylinder, spaced radially 90 degrees apart and 9 in. from the top for the 30-in. BTD and 6 in. from the top of the 24-in. BTD. These four positions are to be labeled front, left, right and rear.

Testing requires the BTD to have pressure transducers positioned at the ear location for auditory evaluation and the thorax position to evaluate non-auditory hazards to air filled organs. BTD chest measurements and accompanying BOP HHA software are used to evaluate non-auditory hazards while ear measurements and AHAAH model are used to assess auditory hazards.

Piezo-resistive transducers are the preferred transducer to measure blast loading in the thorax position due to their ability to measure pressure over a longer period of-time. Piezo-electric transducers are the preferred transducer to measure blast loading the auditory position due to their ability to accurately measure peak pressure and higher frequencies.

The ear gauge is held in the desired location using a piece of flexible metal known as a “gooseneck” and a Delrin gauge adapter (see Figure 24). Shown in Figure 4, the gauge adapter is secured to the gooseneck which is attached to the top stiffener plate, to allow for minor position adjustment. The final height and distance between the chest and ear gauge are listed further below in this document.

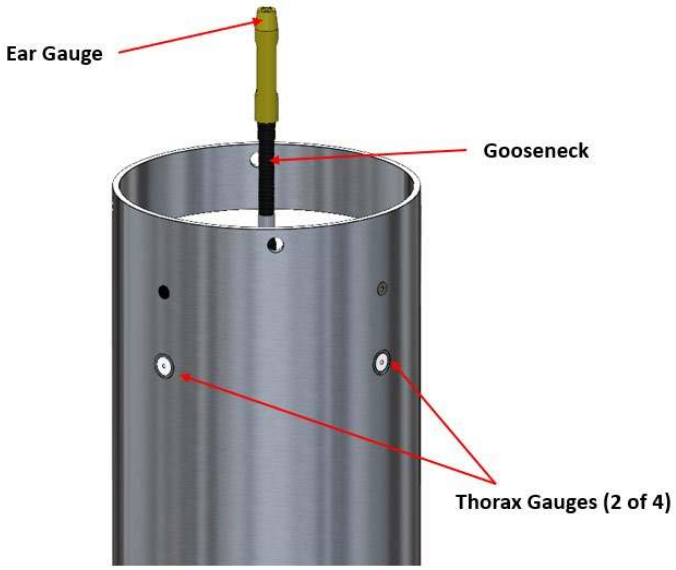


Figure 4 - Chest and Ear Gauges of BTD

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Four BTDs are shown in Figure 5, fully assembled and ready for an enclosed shoulder fired test. As seen in the photo, not all BTDs have the ear gauge assembly, which is an optional attachment used for auditory hazard identification only. Also seen in Figure 5, blue isolating rubber is used underneath the tripod mounting assembly that the BTD attaches to, helping prevent ground shock from ringing the pressure gauges.



Figure 5 – BTD's Positioned in Test Enclosure

The bottom internal stiffener is slotted to allow the top stiffener of the BTD to attach to a tripod or other rigid stand. This configuration yields the most rigid mounting of the BTD which is critical during testing. Ground shock and accelerative loading from testing can cause erroneous results from pressure transducers, therefore rigid mounting, and shock isolation of the BTD and transducers is critical.



Figure 6 - BTD Mounting to Stand

Thorax pressure gauges are installed in specially designed mounts that protect the transducers from mechanical shock and acceleration associated with a blast event. Transducers are mounted in a

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gauge adapter that sits inside a threaded insert which threads into the outside of the BTD tube while being cushioned from direct metal-to-metal contact with Isodamp washers, see Figure 7. Additional images of the transducer mounting assembly can be seen in Figures 8 and 9.

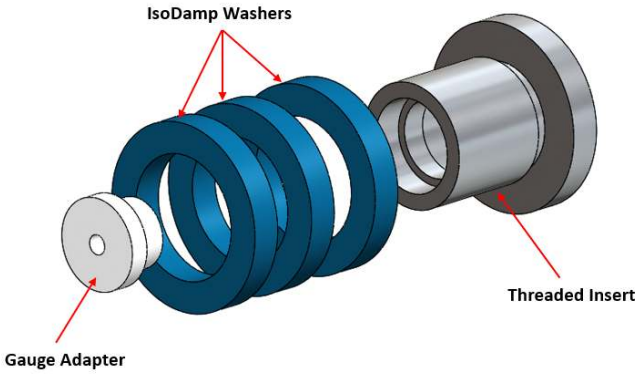


Figure 7 - Transducer Mounting Assembly



Figure 8 – Internal View (with bottom stiffener plate removed) of BTD Thorax Pressure Gauge Mounting



Figure 9 – Complete Transducer Mounting Assembly

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The face of the resulting transducer assembly should be flush with the outer surface of the BTD once installed. To insulate the pressure sensor from flash temperatures of the blast, thermal protection is used. A 1 millimeter (0.04 in.) thick layer of grease on the diaphragm is the preferred method of thermal protection for the transducers. The specific recipe for the grease mixture can be found in Appendix C. In addition to a mounting hole that secures the tripod and gooseneck adapter, the top stiffening plate may have additional holes to allow for transducer cables to run through the BTD.

4.4 Measurements

Measurements for the heights of the BTD transducers comes from Natick technical report TR-15/007, which was a 2012 Anthropometric survey of U.S. Army personnel. Typical Blast Overpressure (BOP) testing is planned to model a 50th percentile male and heights of the BTD and BTD sensor locations should mirror this report. Additionally, these measurements can be helpful when positioning the BTD in non-standing positions. Useful dimensions pulled from TR-15/007 are listed below in Table 1.

Table 1 – 50th Percentile Male Dimensions per Natick TR-15/007

Dimension	Inch	Centimeter
Height (stature)	69.09	175.5
Eye Height	64.57	164
Acromial Height (Shoulder/Top of Torso)	56.65	143.90
Waist Height (Middle of Torso)	41.54	105.5
Crotch Height (Bottom of Torso)	33.27	84.50
Chest Height	50.75	128.90

Note that the distance from ear to shoulder is 20.1 centimeter (cm) (7.92 in.). More importantly the measurement from the Chest to the Ear is 35.1 cm (13.18 in.) which should be met.

The measurement from the Chest and Ear was determined by subtracting the Chest Height and Tragion distances from the Stature value. The stature of the 50th percentile male is 69.09 in., minus chest height of 50.75 in., minus Tragion distance of 5.16 in, yields 13.18 in. This is the distance that the BTD ear transducer should be from the BTD chest transducers.

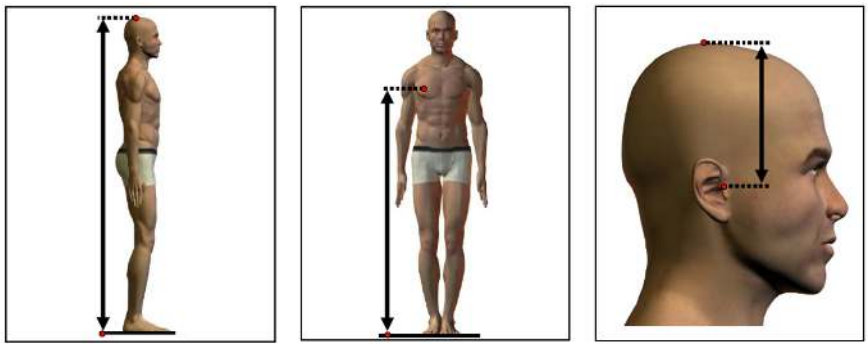


Figure 10 – Determination of Ear Height per Natick TR-15/007

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Typical auditory and non-auditory (BTD) transducer locations are shown below in Figure 11. The Figure also shows the approximation of the human torso for the 24-in. BTD.

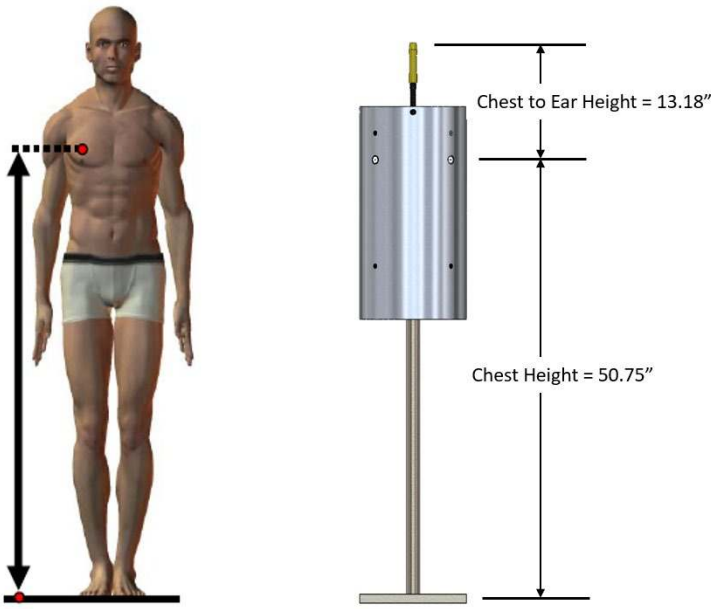


Figure 11 - Anthropomorphic comparison to BTD

5. TEST PROCEDURES.

5.1 Range Setup

Tests are to be conducted on an appropriate firing range for the desired test scenario. Unless otherwise stipulated, all weapons are to be positioned under realistic conditions comparable to those employed in combat or training missions. For weapons with various charges (e.g., separately loaded artillery ammunition), the charge that produces the highest peak pressure level will be measured in addition to those charges producing lower levels.

General measurements are taken around the weapon in the firing position. The firing position shall be free of any extraneous structures. Impulse noise measurements for all weapons are to be recorded at the normal firing elevation. Handguns and shoulder-fired weapons are preferably mounted to minimize reflecting surfaces, with the weapon being placed at the normal firing height, i.e., standing, kneeling, or prone position.

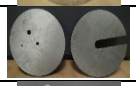
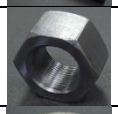
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5.2 BTD Setup

Below will be the assembly and setup procedures for a typical BTD test, using the ATC 24-in. BTD.

1) Gather necessary items to assemble BTD. A breakout of all the necessary parts is shown in Table 2 below.

Table 2 – Parts Required to Build BTD

ITEM	Quantity	Illustration
Cylinder Body	1	
Internal Top and Bottom Stiffener Plate	2	
¼ in.-20 flat head screw	8	
Gage Holder	4	
Threaded Adapter	4	
ISODAMP Washer	4	
O-Ring	4	
Gooseneck	1	
Mounting Stud	1	
Mounting Stud Nut	1	
Mounting Stud Washer	1	
Microphone Adapter	1	
Tripod/Stand Mount	1	

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2) Start by placing the body of the BTD on a table and install the four threaded adapters into the BTD body. Place 3-4 Isodamp washers on the threaded body (Figure 12) and screw into the body, from the inside) until snug (Figure 13). Note the face of the threaded body should be flush with the outside face of the BTD (Figure 14).



Figure 12 – Isodamp Washers Installed on Threaded Adapter



Figure 13 – Threaded Adapter Installed Inside of BTD



Figure 14 – Threaded Adapter Flush with Outside Body

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3) With the threaded adapters in place, insert a transducer through the opening (Figure 15) and screw into the gage holder, using manufactures recommended torque (Figure 16). Ensure O-ring is installed on gauge holder and push into place. Repeat steps 2 and 3 for the remaining three gauges. All transducer assemblies should be flush with the outside of the BTD as shown in Figure 17.



Figure 15 – Transducer Running Through BTD



Figure 16 – Transducer Installed into Gauge Mount



Figure 17 – Final Transducer Placement in BTD

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4) Once the transducers are installed, install the top stiffener plate using 4x ¼ in.-20 flat head screws.
5) Setup tripod that the BTD mounts to and install BTD mounting stud (Figure 18).



Figure 18 – Tripod used to Mount BTD

6) Place BTD assembly on top of the mounting stud. Use caution not to snag any of the transducer cabling when placing the BTD on the tripod. Secure washer and nut on top of BTD to the mounting stud.



Figure 19 – BTD Placed on Tripod

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7) With the BTD on the tripod, tape the transducer cables to the inside wall to provide strain relief (Figure 20).



Figure 20 – Internal Cable Strain Relief

8) In preparation for bottom plate installation, bundle the four transducer cables and tape them to the inside wall of the BTD as shown in Figure 21. This will allow you to slide the cables through the slot of the bottom plate.

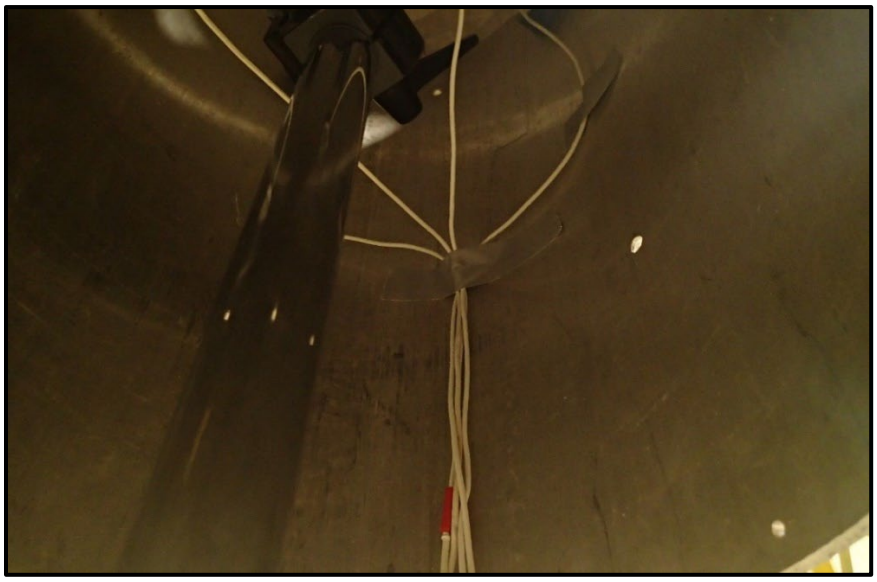


Figure 21 – Tape Preparation for Bottom Stiffener Plate

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9) Install the bottom plate using 4x ¼ in.-20 flat head screws. Use caution not to pinch the transducer cables when installing the plate.



Figure 22 – Bottom Plate Installed

10) If the test calls for auditory measurements prep the microphone to be used by running the correct microdot cable through the microphone adapter as shown in Figure 23. Connect the microdot cable to the auditory pressure gauge and then thread it into the mount.

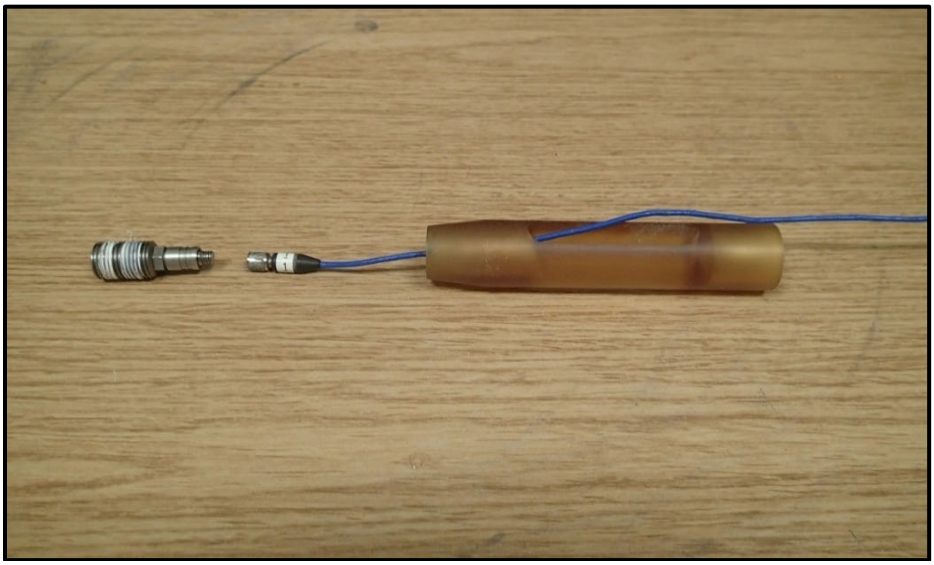


Figure 213 - Installing Auditory Pressure Gauge Cable

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11) Thread the gooseneck adapter to the top of the threaded BTD adapter stud installed in step 5. Then thread the gauge adapter with auditory pressure gauge on to the gooseneck. Run the microdot cable through the cable hole in the top plate.



Figure 24 – Completed Auditory Pressure Gauge Assembly

Repeat steps 1-11 for any remaining BTDs that are needed for the required test. At this point they are ready to be transported to the test range. Secure loose cabling and protect transducers so they are not damaged in transport.

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5.3 Test Execution

Position BTDs in desired location that best simulate the desired weapon/threat scenario. Record measurements and photograph the position of each BTD so that it can be included in the test report. Ensure the correct BTD height is being used for the selected test scenario.

In Figure 25 below, two BTDs are set up for a shoulder fired weapon. Between the two BTDs is a weapon stand that was designed and fabricated at ATC specifically for Noise and BOP measurements. It has adjustable height and minimal reflective surfaces that allow the BTD's to fit snugly against the weapon being tested. The two BTDs represent the Gunner and the Assistant Gunner.



Figure 25 – BTD Test Setup

Ensure that the BTD thorax gauges are aimed in a manner which simulate how the subject would be positioned during the test scenario.

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6. DATA REQUIRED

The following data is required:

a. Complete list of instrumentation used for the measurements, including name, type, serial number, manufacturer, and calibration information.
b. Complete identification of the weapon system.
c. Complete description of the operational conditions under which measurements were made, such as transducer locations and orientation, weapon height and elevation, muzzle velocity, hatch position, etc.
d. Date, time of day, and location of test.
e. Physical description of the area, including the composition of the ground surface, the location of the reflecting surfaces, and a sketch of the weapon and transducer and microphone locations, as required.
f. Temperature, relative humidity, wind direction and velocity, and atmospheric pressure, as required. The recorded temperature and pressure may be used to compute peak pressure, duration and impulse data, scaled to standard temperature and pressure conditions, as required.
g. Type and lot number of projectiles, propellants, and/or explosives.

7. PRESENTATION OF DATA.

Provided below is an example of a summary report that can be used to provide emerging results to test customers. Additional detailed requirements for the presentation of test data should be coordinated with the IPT on a program-by-program basis and reflected in the test plan(s) as appropriate.

NOTIONAL EXAMPLE OF BLAST OVERPRESSURE TEST REPORT

2.3 Blast Overpressure Report.

2.3.1 Objectives.

The objective of this testing is to determine the blast-overpressure and noise levels at various crew positions while firing the XXX weapon.

2.3.2 Criteria.

While firing the XXX weapon, the combination of blast overpressure and duration at the crew positions shall not exceed the impulse-noise limit, Category Z, as defined in MIL-STD-1474D (Test Agency devised, ATEC approved).

2.3.3 Test Procedure.

a. A total of YY rounds were fired from the XXX weapon. The rounds were fired from X through

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XX July 2XXX at Aberdeen Proving Ground, Maryland. The rounds were temperature conditioned to –XX °F (-YY °C), and were fired from the XXX firing position. An IR detector was used to obtain the t-zero time for these test firings.

b. PCB and Endevco type pressure transducers were used to measure the blast overpressure levels. Measurements were taken at the gunner and assistant gunner positions. Blast Test Devices (BTDs) were used at the gunner and assistant gunner positions. Data from the PCB auditory transducers were sampled at a fixed sampling rate of two hundred thousand samples per second per channel and recorded digitally with a forty kilohertz low pass filter, while the Endevco transducers in the BTD (non-auditory locations) were sampled at a fixed sampling rate of two hundred thousand samples per second per channel and recorded digitally with a ten kilohertz low pass filter.

c. Measurements for the gunner's head and chest positions were derived after careful consideration of the physical features of the weapon and measurements for the fiftieth percentile soldier as presented in the 2012 Anthropometric Survey of U.S. Army Personnel: Methods and Summary Statistics.

d. The data was acquired using the Ballistics Instrumentation Divisions data acquisition system. The system provides for signal conditioning (analog amplification and filtering) of various transducers and digital conversion and storage of the data. Personnel of the Ballistics Instrumentation Division, Live Fire Directorate performed the data acquisition.

2.3.4 Test Results.

a. Ambient atmospheric pressure and temperature data values were averaged over the firing times on each of the firing days. The data presented in this report have not been corrected to standard atmospheric conditions because of the close proximity of the firing site to sea level.

b. The peak side-on blast overpressure Pa, the A and B duration times, the allowable number of rounds per day, NB, and the impulse were recorded for each test round fired. The peak pressure levels were taken in kilopascals (kPa) and converted to decibels (dB) using the formula: dB = 153.98 + 20 log (kPa). The mean and worst-case condition for each group of rounds fired was also calculated. The round-by-round blast overpressure data is available from the Analytical Team upon request.

2.3.5 Technical Assessment.

a. The allowable number of rounds per day that can be fired with single hearing protection, in accordance with MIL-STD-1474D, was less than five rounds per day.

b. The non-auditory blast overpressure data obtained at the chest positions for the gunner and assistant gunner from this test will be forwarded to Defense Centers for Public Health-Aberdeen (DCPH-A) for further auditory and non-auditory data analysis.

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Summary Table of Blast Test Device (BTD) and ***Auditory Data (Shown in Bold)***

Date Fired		Round Type			Firing Position			Cond. Temp			
		Location									
		Radial from Center of Weapon, degrees	Distance from front of weapon, cm	Distance from center axis of weapon, cm	Height Relative to weapon, cm	Peak Pressure, kPA		B Durtaion Time, ms		[b]ANOR/Day	
Channel Number	Crew Position					Mean	[a]Worst Case	Mean	Worst Case	Mean	Worst Case
1	Gunner (Front)	N/A	72	20.3	-6.6	13	15	15	14	57	24
2	Gunner (Right)	N/A	72	20.3	-6.6	15	18	15	15	32	12
3	Gunner (Rear)	N/A	72	20.3	-6.6	36	50	11	10	1	0
4	Gunner (Left)	N/A	72	20.3	-6.6	21	31	13	11	13	2
5	Gunner (Head)	N/A	58.4	16.7	6.6	24	23	45	76	1[d]	0[d]
6	Asst Gunner (Head)	N/A	50	42.1	2.3	20	21	18	17	6	5
7	Asst Gunner (Front)	N/A	46.6	58.4	-17.1	21	25	12	12	9	4
8	Asst Gunner (Right)	N/A	46.6	58.4	-17.1	14	15	15	15	37	26
9	Asst Gunner (Rear)	N/A	46.6	58.4	-17.1	20	24	14	12	9	4
10	Asst Gunner (Left)	N/A	46.6	58.4	-17.1	38	42	11	10	0	0

Summary Table Notes:
a Worst case value is P3.0103 * t.
b ANOR/day is NB, the allowable number of rounds per day with single hearing protection.
c Exceeds one thousand rounds per day.
d Data exceeds the Z-curve. Prior approval by the Surgeon General's Office is required.
e Height above ground.
f Distance from center of weapon along the radial.

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APPENDIX A. GLOSSARY.

Term	Definition
Blast Overpressure	A term used to describe the sudden release of gases into the surrounding air from the detonation of an explosive or the firing of a gun. This explosion causes a shock wave or front to be propagated outward from the source.
Blast Test Device (BTD)	A cylindrical fixture that simulates the human torso in approximate volume. BOP gages are assembled into the BTD to assess the effect of blast overpressure on Army Personnel working within or around Army Materiel.
Piezoelectric Pressure Transducer	A device using the piezoelectric effect to convert an observed pressure into a specific electrical charge.
Piezo-Resistive Pressure Transducer	A device using the piezo-resistive effect to convert an observed pressure into a change of resistance as a result of deformation of a resistance leg in a Wheatstone Bridge.
Shock Wave	A shock wave is a type of propagating disturbance. Shock waves are characterized by an abrupt, nearly discontinuous change in the characteristics of the medium through which it passes. Across a shock there is always an extremely rapid rise in pressure, temperature and density of the flow.
Gage Factor or Gage Sensitivity	The ratio of observed pressure to the electrical signal produced; provided on Gage Calibration Certificate. Gage Sensitivity is the inverse of Gage Factor (i.e., V/psi vs. psi/V).

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APPENDIX B. ABBREVIATIONS.

ATC	Aberdeen Test Center
ATEC	Army Test and Evaluation Command
ANOR	Allowable Number of Rounds Per Day
BOP	Blast Overpressure
BOP-HHA	Blast Overpressure Health Hazard Assessment
BTD	Blast Test Device
cm	Centimeter
dB	Decibels
DCPH-A	Defense Centers for Public Health-Aberdeen
FSO	Full Scale Output
ft	Feet
Inch(es)	in.
kHz	Kilohertz
m	Meters
MIL-STD	Military Standard
NIST	National Institute for Standards and Technology
Pa	Pascal
TOP	Test Operations Procedure
µs	Microseconds
°	Degrees

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APPENDIX C. GREASE RECIPE

Mica Powder, Phlogopite (Kish Co Inc. HR2 type)
(http://www.kishcompany.com/index.php/micas/thermaflex-phlogopite-mica)

- Phlogopite is heat resistant, does not transmit electricity, and is a poor conductor of heat.

Dow Corning Hi-Vacuum Silica Synthetic Grease, 5.3 oz. (150 gm) Tube (McMaster-Carr P/N 2966K52)

- Binding agent that gives stability to the compound.

Black Toner Cartridge powder

- Inhibits light penetration that creates light contamination.

MIXTURE:

Three parts Mica Powder.
Two parts Silica Synthetic Grease.
One-part Black Toner Cartridge powder.

The final mixture consistency should be slightly thicker than toothpaste. It is recommended to mix the container prior to application in the event any separation has occurred during long term storage.

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APPENDIX D. REFERENCES.

1. US MIL-STD-1474E, Department of Defense Design Criteria Standard, Noise Limits, April 2015

2. Technical Report NATICK/TR-15/007, Anthropometric Survey of U.S. Army Personnel: Methods and Summary Statistics, December 2014.

3. FR/GE/US International Test Operations Procedure (ITOP) 4-2-822, Electronic Measurement of Airblast Overpressure, 25 September 2000.

4. MacFadden L, Chan PC, Ho K, Stuhmiller JH: A model for predicting primary blast lung injury. J Trauma Acute Care Surg 2012; 73(5):1121–9.

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APPENDIX E. APPROVAL AUTHORITY.

DEPARTMENT OF THE ARMY
UNITED STATES ARMY TEST AND EVALUATION COMMAND
6617 ABERDEEN BOULEVARD, BUILDING 2202–THIRD FLOOR
ABERDEEN PROVING GROUND, MD 21005-5001

CSTE-DCG-DT (73-1f)

23 JUL 2025

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Test Operations Procedure 04-2-831A USE OF BLAST TEST DEVICE (BTD) DURING BLAST OVERPRESSURE MEASUREMENT Publication

1. Test Operations Procedure (TOP) 04-2-831A USE OF BLAST TEST DEVICE (BTD) DURING BLAST OVERPRESSURE MEASUREMENT has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. This TOP provides guidance on the use of Blast Test Devices in blast overpressure testing. A BTD is a rigid cylindrical test apparatus approximating a human torso and can be used in blast overpressure tests to record the auditory and non-auditory health hazards of a weapon or explosive.

3. This document is approved for publication and will be posted to the ASSIST website. The ASSIST website can be accessed at https://assist.dla.mil/.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army White Sand Missile Range (CSTE-DCG-DT), 100 Headquarters Avenue, White Sands Missile Range, NM 88002-5000; or e-mailed to usarmy.apg.atec.mbx.atec-standards@army.mil.

ANDREW R. MORGAN
COL, FA40C
Deputy Commanding Officer
for Developmental Testing

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